Numerical Methods

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Euler's method

$$\begin{cases} \frac{dy}{dt} = f(t,y) \\ y(t_0) = y_0 \end{cases}$$

soln: $y = \phi(t)$

approximation of $y = \phi(t)$ at $t = t_{n+1}$:

$$y_{n+1} = y_n + f(t_n, y_n)(t_{n+1} - t_n)$$

linear approximation on interval $[t_n, t_{n+1}]$:

$$y(t) = y_n + f(t_n, y_n)(t - t_n)$$

special case: uniform step size

$$y_{n+1} = y_n + hf(t_n, y_n)$$

Ex. $\frac{dy}{dt} + \frac{1}{2}y = \frac{3}{2} - t$ $y(0)=1$ $0 \leq t \leq 1$, $h = 0.2$

$f(t,y) = -\frac{1}{2}y - t + \frac{3}{2}$

$f_0 = f(0,1) = -0.5 - 0 + 1.5 = 1$

$\therefore$ for $0 \leq t \leq 0.2$,

$y \approx 1 + 1.0(t-0) = 1 + t$

$\Rightarrow y_1 = y(0.2) = 1.2$

$f_1 = f(0.2, 1.2) = -0.6 - 0.2 + 1.5 = 0.7$

$y_2 = 1.2 + (0.7)(t - 0.2)$

$y_2 = 1.06 + 0.7t$

$\Rightarrow y_3 = y(0.4) = 1.34$

$f_2 = f(0.4, 1.34) = \sim$

Accuracy of numerical methods

Global truncation error: $E_n = \underbrace{\phi(t_n)}_{\text{soln.}} - \underbrace{y_n}_{\text{approx.}}$ error in many steps

① approximate formula for y_{n+1}

② two input data are only approximately correct $(\phi(t_n) \neq y_n)$

Local truncation error: if assume $\phi(t_n) = y_n$ error in any given step

Suppose ϕ is a soln of IVP and f, f_t, f_y are continuous.

$\phi'(t) = f(t, \phi(t))$

$\phi''(t) = f_t(t, \phi(t)) + f_y(t, \phi(t))(\phi'(t))$

$= f_t(t, \phi(t)) + f_y(t, \phi(t)) f(t, \phi(t))$

$t_n < \bar{t}_n < t_n + h$

Taylor polynomial: $\phi(t_n + h) = \phi(t_n) + \phi'(t_n)h + \frac{1}{2}\phi''(\bar{t}_n)h^2$

$\phi(t_{n+1}) - y_{n+1} = \underline{[\phi(t_n) - y_n]} + h\{\underline{f[t_n, \phi(t_n)] - f(t_n, y_n)}\} + \frac{1}{2}\phi''(\bar{t}_n)h^2$

LTE: Take $\underline{y_n = \phi(t_n)}$ $\Rightarrow T_{n+1} = \phi(t_{n+1}) - y_{n+1} = \frac{1}{2}\phi''(\bar{t}_n)\underline{h^2}$

$\boxed{T_n \leq \frac{Mh^2}{2}}$, where M is max of $|\phi''(t)|$ on interval $[a,b]$

$\boxed{T_n \sim O(h^2)}$ (Local truncation error on the order of h^2)

GTE: over $N = \frac{T - t_0}{h}$ steps, $e_n \sim N \cdot O(h^2) = \frac{T - t_0}{h} O(h^2) \sim (T - t_0) O(h)$

$\boxed{e_n \sim O(h)}$

Theorem: If $\underset{\text{local}}{T_n} \sim O(h^p)$, then $\underset{\text{global}}{e_n} \sim O(h^{p-1})$

Lipschitz continuity: $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$

Lipschitz constant, smallest L to make inequality true

$y' = f(t,y)$

derivative is not too big (not unbounded)

If f is Lipschitz, then $|e_n| \leq \frac{\max_j T_j}{hL}(e^{L(T - t_0)} - 1)$

GTE in terms of LTE

Ex. $y' = y$, $y(0) = 1$

↳ $\varphi(t) = e^t$, $\varphi(t_n) = e^{t_n}$

↳ $y_{n+1} = y_n + hy_n$

$= \underline{(1+h)y_n}$

$= \underline{(1+h)^{n+1}} y_0$

repeat this n+1 times

↳ $y_n = (1+h)^n$

$e_n = \varphi(t_n) - y_n = e^{t_n} - (1 + \underline{h})^n$

good for nonzero h, but
h→0 will make error go exponential

Improved Euler's Method

$$\begin{cases} y' = f(t,y) \\ y(0) = y_0 \end{cases} \Rightarrow \varphi(t_{n+1}) = \varphi(t_n) + \frac{1}{2}(t_{n+1} - t_n)(f(t_n, y_n) + f(t_{n+1}, y_{n+1}))$$

$y(t_{n+1}) = y(t_n) + hf(t_n, y(t_n)) + \frac{h^2}{2}\frac{\partial f}{\partial t}(t_n, y(t_n)) + \frac{h^2}{2}\frac{\partial f}{\partial y}(t_n, y(t_n)) + \ldots + O(h^3)$

$y_n^* = y_n + hf(t_n, y_n)$

$\underline{f(t_{n+1}, y_n^*)} = f(t_{n+1}, y_n + hf(t_n, y_n))$

$y_{n+1} = y_n + \frac{h}{2}(f(t_n, y_n) + \underline{f(t_{n+1}, y_n^*)})$

LTE $\sim O(h^3)$

GTE $\sim O(h^2)$

Runge - Kutta

RK(2): $k_1 = f(t_n, y_n)$

$k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$

$y_{n+1} = y_n + hk_2$

LTE: $O(h^3)$ GTE: $O(h^2)$

RK(4): $k_1 = f(t_n, y_n)$

$k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$

$k_3 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$

$k_4 = f(t_n + h, y_n + \frac{h}{2}k_3)$

$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$